

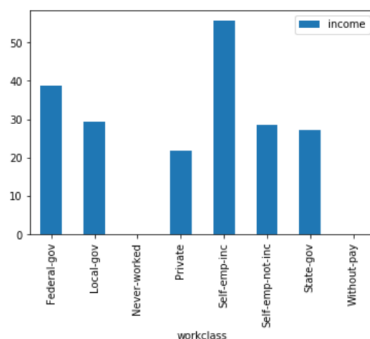
## Problem Statement

This project is about developing marketing profiles for UVW College using data from the US Census Bureau to answer a customer's question focusing on a \$50,000 salary as the key demographic. The marketing team is to predict the income of an individual based on different values of the input parameters so that they can tailor their marketing efforts when reaching out to the individuals based on different factors.

## Progress and Background Work

**Background Work.** Throughout the course, I gained comprehensive understanding on different visualizations, and its implementation on python programming using jupyter notebook. I learned about different tools for data exploration and developed skills using them. Gathered knowledge about Data transformations, Exploratory querying, Statistical graphics. Having a good understanding of visualization and graphs, I started working with databases using sqlite. In addition, I also learned supervised and unsupervised learning, temporal analysis, time series modeling, and how to visualize temporal data and learned about a variety of data exploration techniques such as strings and sequences, edit distance, filtering, and more. I applied this knowledge on different visualization exercises where I had plotted bar charts, scatter plot, pie chart, time series graph etc. Now I have a good understanding of when it is most appropriate to use them in order to present and analyze specific kinds of data.

**Progress.** To report the progress made so far, I understood the input description of the problem. Reviewed the dataset and descriptions of the attributes. After familiarizing myself with the data I performed required data cleaning to address the "?" values in the dataset. Furthermore, I Answered the questions in the Required Checkpoint quiz. In order to complete the project, I have begun to consider various scenarios that would be appropriate for this problem statement. Some of the scenarios that I came up with are given below. In the

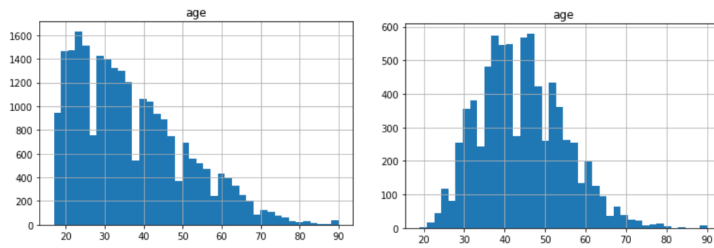


**Scenario 1:** The marketing department wants to know how workclass is related to income.

Fig. 1 describes the relationship between workclass and income.

From the figure, we find more than 50% of the self-emp\_inc workclass get paid above 50k.

**Scenario 2:** UVM marketing department wants to know the relationship between age and income.



The two histogram graphs in Figure 2 provide insight into the relationship between age and income. The first graph showcases the distribution of individuals who earn less than \$50,000 are mostly young people, while the second graph presents the age pattern for

those who earn more than \$50,000 are mostly middle age. By comparing these two graphs, we can observe the clear correlation between age and salary.

## Completed Tasks

The works that I have already completed are reported as follows:

- I acquired a background on different visualization techniques.
- I solved weekly assignments on visualization.
- I gained the understanding of the problem and cleaned my dataset where it was required.
- I started exploring scenarios and using visualization to determine the most suitable approach for each scenario.

## Challenges

- Dataset cleaning: I faced challenges while cleaning the dataset using pandas, because the attributes are string type.
- Pandas grouping: I also faced technical difficulties due to pandas group by functionality. After grouping, I needed to apply `reset_index` to convert the data.
- Plot type in age vs income: Another challenge I encountered was in plotting the relationship between age vs income in a line chart as they both are numerical data. Instead I utilized two different histograms for different incomes to portrait the relationship between age and income.

## Work to Complete and Plan for Completing

- Moving forward, I plan to consider additional scenarios and implement them to find out the best visualization. This will involve creating necessary conclusions for each scenario.
- In addition, I will also complete any remaining course material that may assist in further developing my visualizations. I believe that these additional resources will be crucial in completing my project to the best of my ability.